1. Introduction

Python Programming / Operating Systems

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Scope and Goal

- The basic concepts of programming
- Programming in Python
- The tools you need to program (as far as necessary)
- Basics of computer networks
- Operating systems and shell programming
- Gaining practical experience with all of the above
- Disclaimer: the actual work starts after this course!

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Organization

- We provide you with:
  - Basics of computer networks and Operating systems (the theory)
  - Concepts and tools for Python program development (the theory)
  - Exercises and programming tasks (the practice)

- You need to:
  - Ask questions (at any time) and engage with your classmates! (we are a small group)
  - Work through the tasks (on your device)

- Course Material:
  - Moodle
  - Slides
  - Lecture Notes

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Content

Python Programming

- Foundation of Programming
- Primitives (Data Types)
- Operations
- Collections
- Control Statements
- Functions
- Classes
- Threads
- Docker

Basics of Computer Networks

- Network Protocols
- ISO/OSI Model
- TCP/IP Model

Operating Systems (OS)

- Binary Code
- Computer Architecture
- Fundamentals of OS
- OS services and functions
- Linux
  - Shell Programming

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1.1 Getting Started

- Python Program
- Environment

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Python Program

Python

Program

Problem

Programming Language

Algorithm

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Problem 1/2

Description:
A **task** or **question** that requires a solution or answer is a **type of problem** that challenges you to think critically and apply your knowledge to solve it. This can range from a simple question to more complex problems that require **multiple steps** to find a solution.

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Problem 2/2

Example:
Calculate the mean of some given numbers. For instance, if you are given the numbers 4, 8, 15, 16, 23, and 42, you would:

- Add them together: 4+8+15+16+23+42=108.
- Count the total numbers: 6.
- Divide 108 by 6: 108/6=18.

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Algorithm 1/3

Description:

An **abstract step-by-step procedure** to solve a problem is often called an **algorithm**. Formally, it is a defined **sequence of instructions** or a **systematic approach** used to achieve a particular goal or solve a specific problem.

An algorithm is fundamental in both computer science and everyday problem-solving, as it helps **break down complex challenges into manageable steps** that can be followed to reach a solution.

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Algorithm 2/3

A good algorithm is:

- **Clear:** each step is unambiguous, and there is no confusion about how to execute it.
- **Efficient:** it solves the problem in a reasonable amount of time without unnecessary steps.
- **Generalizable:** it can be applied to different sets of data, not just a specific example.

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Algorithm 3/3

Example:
Let's take the example of calculating the mean of a set of numbers, and see how this can be expressed in a step-by-step algorithm:

1. Sum up all numbers.
2. Count all numbers.
3. Divide the sum by the count.

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Program

Description:

A **set of instructions** telling your computer how to execute an algorithm.

Example:

```
sum = 0
count = 0

for number in numbers:
    sum = sum + number
    count = count + 1

mean = sum / count
```

The **programm text** is the **program code**

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Programming Language

Description:
A **specific language for formulating programs** is called a **programming language**. Programming languages are used by developers **to communicate instructions to a computer** in a way that it can understand and execute. They are the medium through which we create software, automate tasks, and solve computational problems.

Example:
Python, Java, or C++

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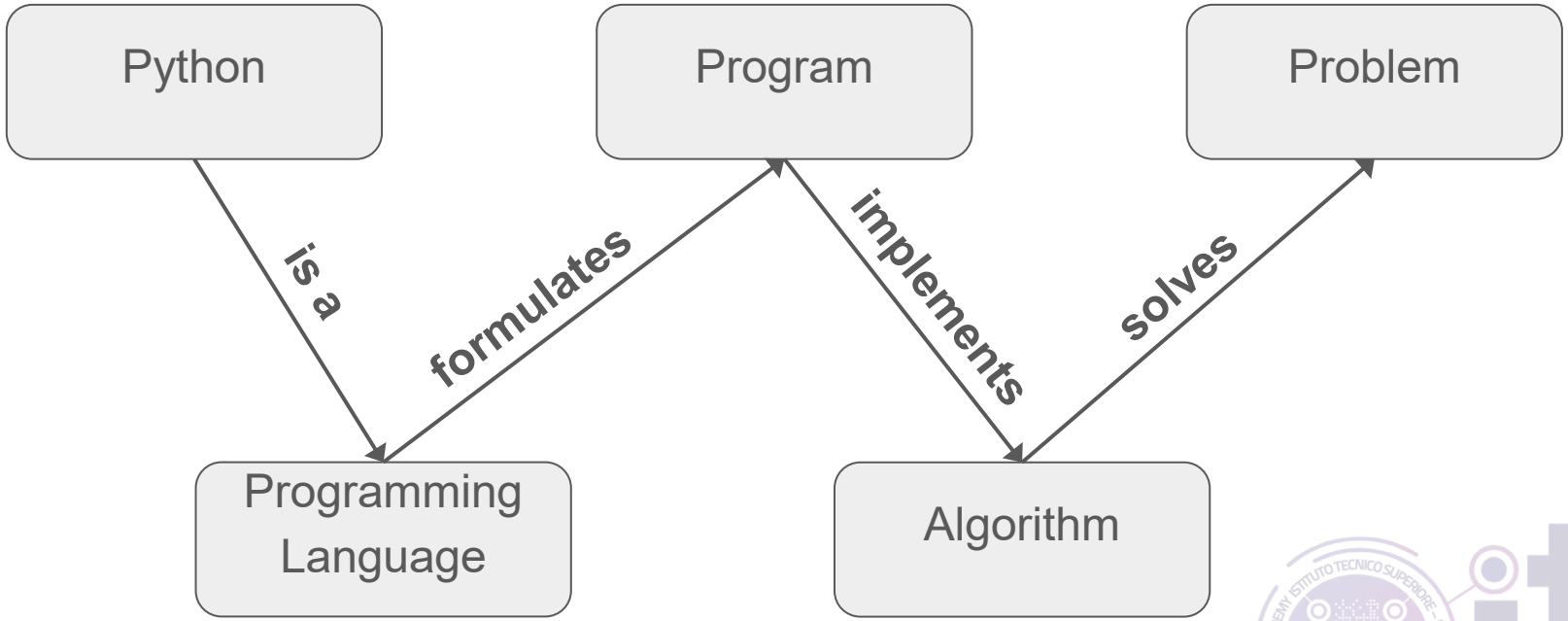
Python Program



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Environment 1/2

Permanent Storage

- Slow to access
- E.g. HDD, SSD, NAS, cloud storage, etc.

Your Data

Python program code

Processing Units

- Do the actual computation
- Central Processing unit (CPU)
- Graphics Processing Unit (GPU)

Temporary Storage

- Fast to access
- Only temporary
- E.g. RAM, VRAM

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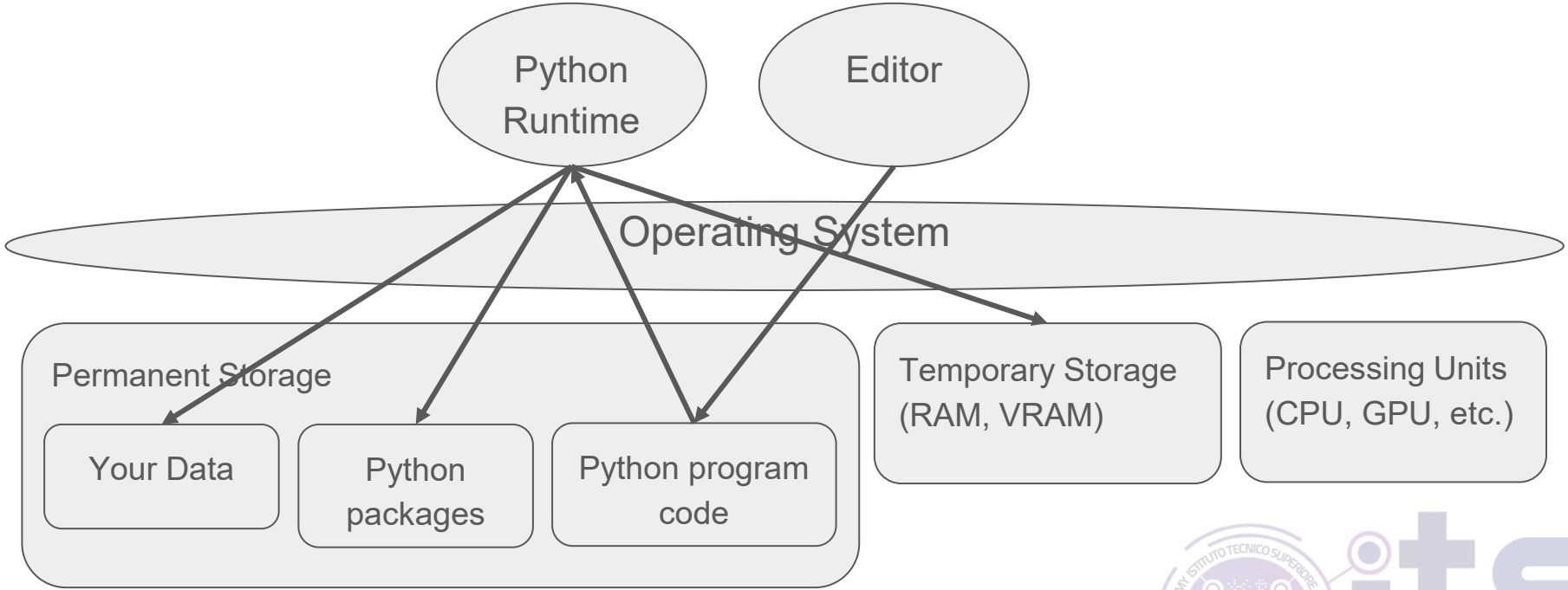
Environment 2/2



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1.2 Setting up our tools: Command Line

- Commands:
  - ls
  - cd
  - mkdir
  - touch
  - rm

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Command-Line

- With the **command-line** (cmd) you can communicate directly with your computer (i.e. operating system)
- Example: For listing the content inside a folder (`ls`)

```
39348@DESKTOP-R63USTJ MINGW64 ~/OneDrive/Desktop
$ ls
'[ML2023] Counterfactual Explainable AI.pptx'     BigliettoRomaFCO-RomaTiburtina.jpeg     dblp/
"AAAI'24 - RSGG-CE.mp4"                           BigliettoRomaFCO-RomaTiburtina.pdf      dblp_condgce/
"AAAI'24 - RSGG-CE.pdf"                          'bollini judo'/                          desktop.ini
 AF1504_2024-02-26_CDG-FCO.pdf                   'Canada Visa'/                           Email_Addresses.xlsx
 AssicurazioneSanitariaRegionale.pdf             'Certificate -        Bardh Prenkaj.png' experiments/
 attendance_certificate_conference-2024.pdf       check_in_details.pdf                    facebook_ct1.zip
'AUTORIZZAZIONE MISSIONE_AAAI_24_PRENKAJ_GS.pdf'  code-wsn.zip                           FISA_2023_proposal.docx
 bae-master/                                      Contracts4TUM/                          FISA_2023_proposal.pdf
 bae-master.zip                                   Contratto_Biter.pdf                     GMT20231129-192311_Recording.transcript.vtt
 Bardh-Prenkaj-2135604751.pdf                     Dataset.zip                             graph_counterfactual_explainers_colored_diagram.tikz

39348@DESKTOP-R63USTJ MINGW64 ~/OneDrive/Desktop
$
```

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Command-Line - Try it yourself

1. Open command-line: (Use the shortcut **ctrl+alt+t**)
2. Go to your desktop: **`cd`** `Desktop`
3. Create now folder: **`mkdir`** `<name of your folder>`
4. Show your Desktop content:
   - Linux: `ls`
5. Remove folder: **`rm -r`** `<name of your folder>`

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Command-line - Navigating directories

- Change your current directory (**working directory**) with `cd <target>`

```
39348@DESKTOP-R63USTJ MINGW64 ~/OneDrive/Desktop
$ cd dblp

39348@DESKTOP-R63USTJ MINGW64 ~/OneDrive/Desktop/dblp
$ mkdir cartella-di-prova
```

- For going up the folder structure use “..” as target

```
39348@DESKTOP-R63USTJ MINGW64 ~/OneDrive/Desktop/dblp
$ cd cartella-di-prova/

39348@DESKTOP-R63USTJ MINGW64 ~/OneDrive/Desktop/dblp/cartella-di-prova
$
```

```
39348@DESKTOP-R63USTJ MINGW64 ~/OneDrive/Desktop/dblp/cartella-di-prova
$ cd ..

39348@DESKTOP-R63USTJ MINGW64 ~/OneDrive/Desktop/dblp
$
```

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Create your first “empty” Python file

You can do that using the command **touch** *<file_name>*

```
39348@DESKTOP-R63USTJ MINGW64 ~/OneDrive/Desktop/dblp/cartella-di-prova
$ touch first_python_program.py

39348@DESKTOP-R63USTJ MINGW64 ~/OneDrive/Desktop/dblp/cartella-di-prova
$ ls
first_python_program.py

39348@DESKTOP-R63USTJ MINGW64 ~/OneDrive/Desktop/dblp/cartella-di-prova
$ 
```

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1.3 Setting up our tools: Integrated Development Environment (IDE)

- IDE
- VSCode

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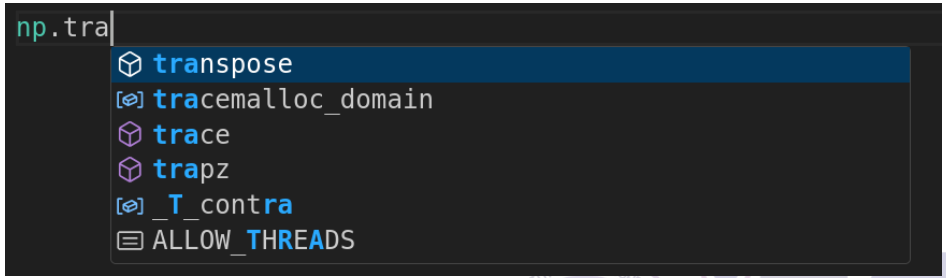
Visual Studio Code - Programming more comfortably!

- **Code editors** are great for developing programs!
  - Visual Studio code is actually more like a *Integrated Development Environment* (IDE)
- Write code and **save** it as a file
- **Highlight errors** in code
- **Autocomplete** and recommendations



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- **Documentation** and references

```
np.transpose()
```

```
(a: _ArrayLike[_SCT@transpose], axes:
_ShapeLike | None = ...) ->
NDArray[_SCT@transpose]
```

Returns an array with axes transposed.

For a 1-D array, this returns an unchanged view of the original array, as a transposed vector is simply the same vector. To convert a 1-D array into a 2-D column vector, an additional dimension must be added, e.g., `np.atleast2d(a).T` achieves this, as does `a[:, np.newaxis]`. For a 2-D array, this is the standard

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- Write code and **save** it as a file
- **Highlight errors** in code
- **Autocomplete** and recommendations
- **Documentation** and references
- Code **versioning** control with Git → more on Git later!

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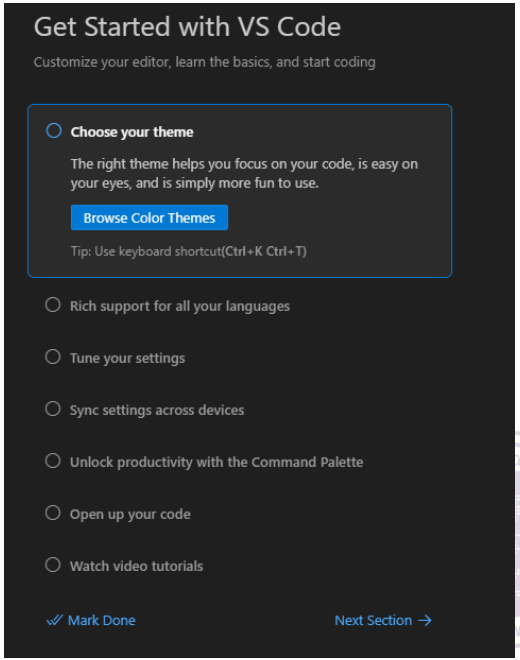
Visual Studio Code - Installation

- Go to code.visualstudio.com
- **Download** installer & **Install** (should be similar for Windows & Mac)
  - Select **all** additional tasks
- **Wait** for the others if you are at "Get Started with VS Code" (We will do the setup together)



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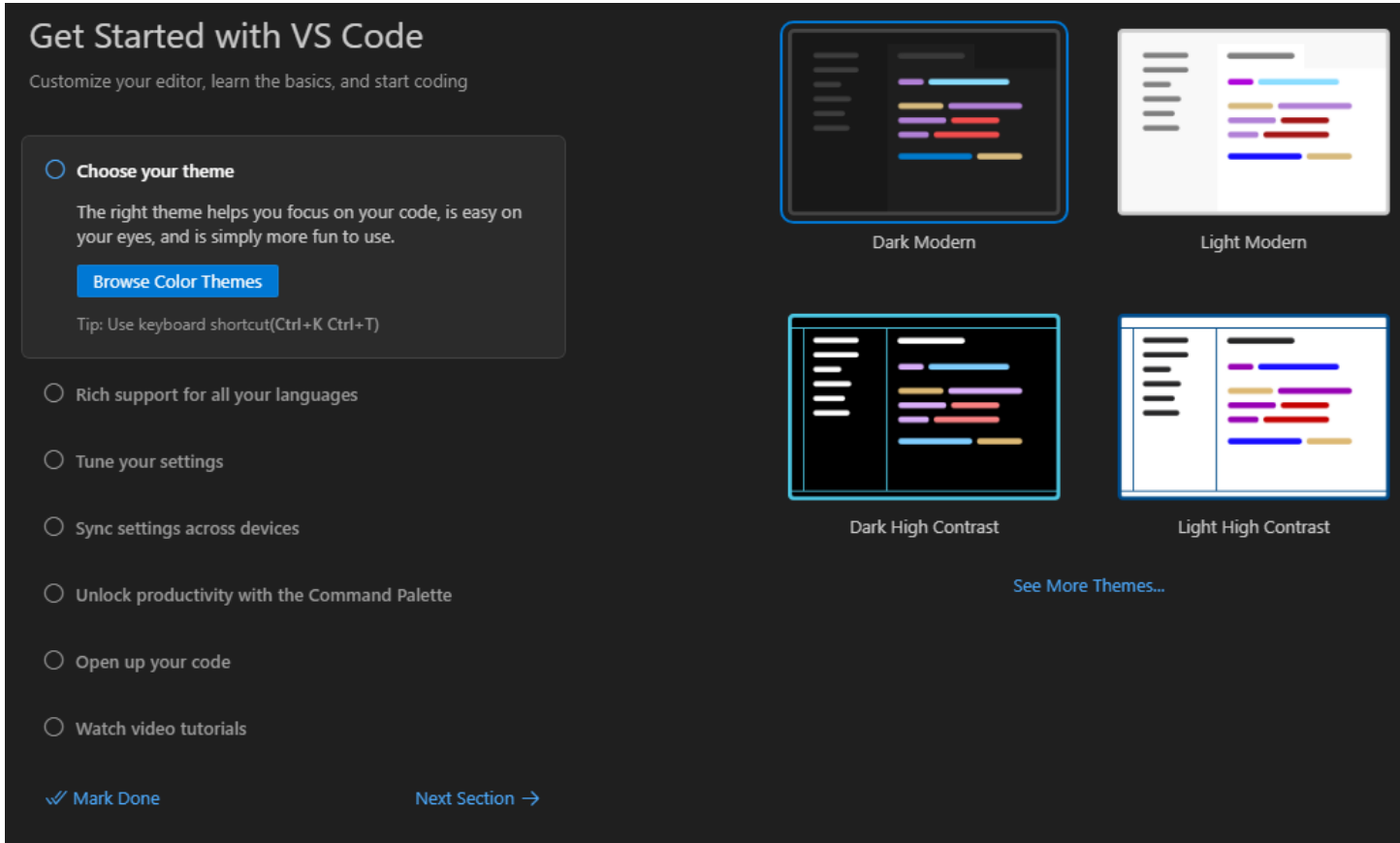
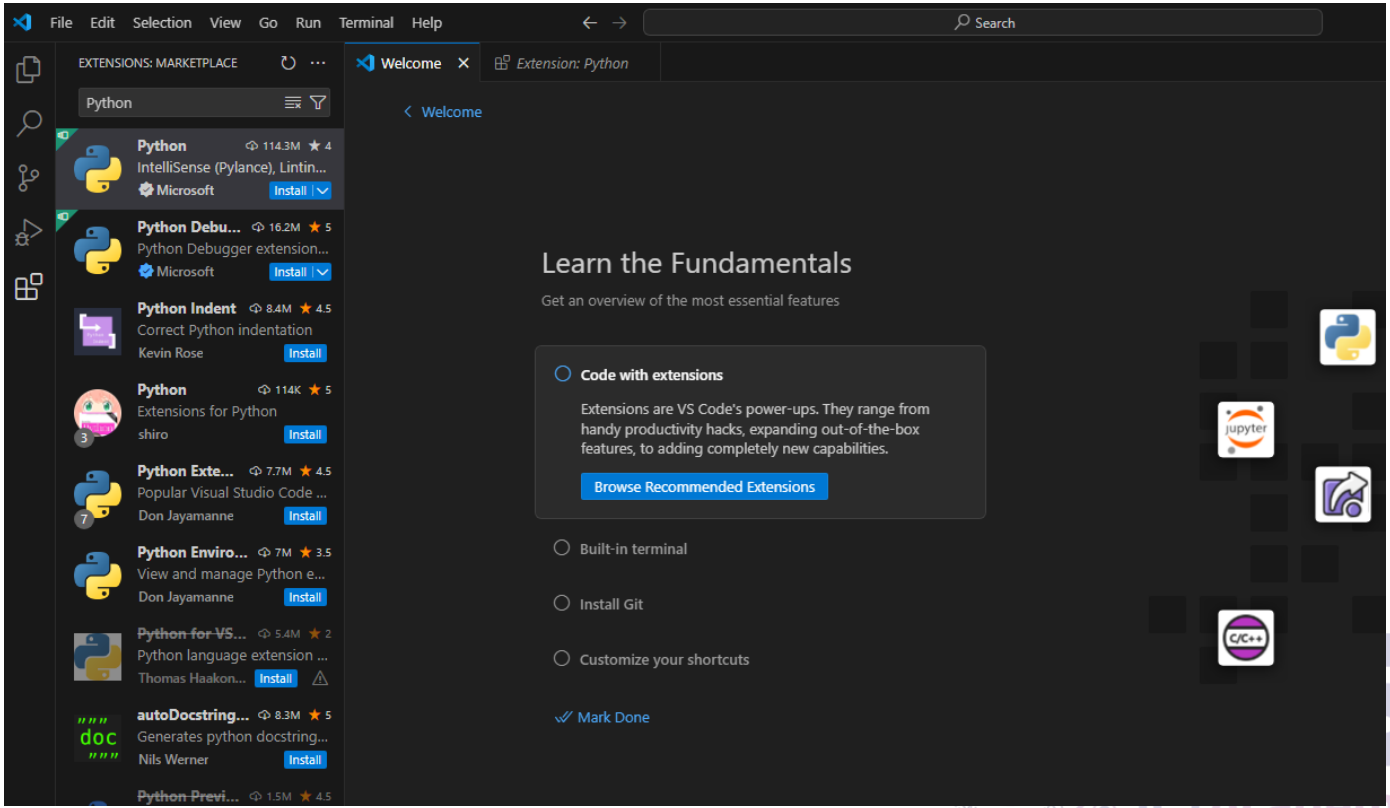
Visual Studio Code - Setup - Theme

1. **Choose** Theme
2. Click “**Next Section**”



Visual Studio Code - Setup - Python Extension

1. Click “**Browse Recommended Extensions**”
2. Type “**Python**” in the extension **search** bar
3. **Install** the “**Python**” extension
4. If you get the “Get Started with Python Development” Tab opens; **wait** for the others



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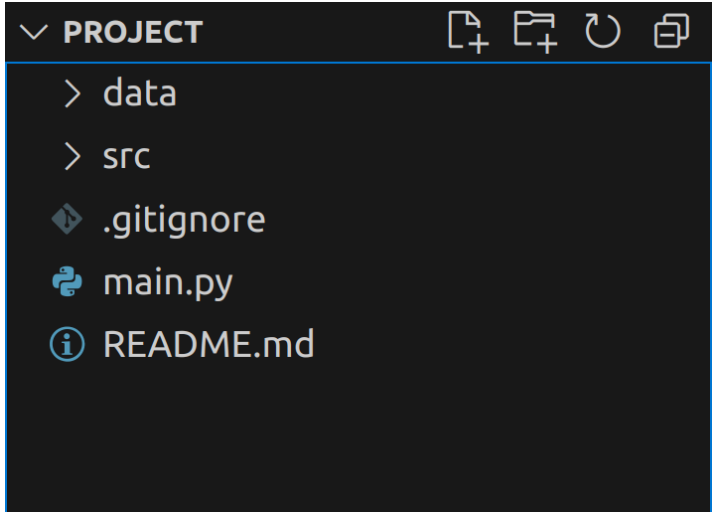
Exercise: Let’s create a “standardized” Python project structure using: mkdir, cd, touch, ls



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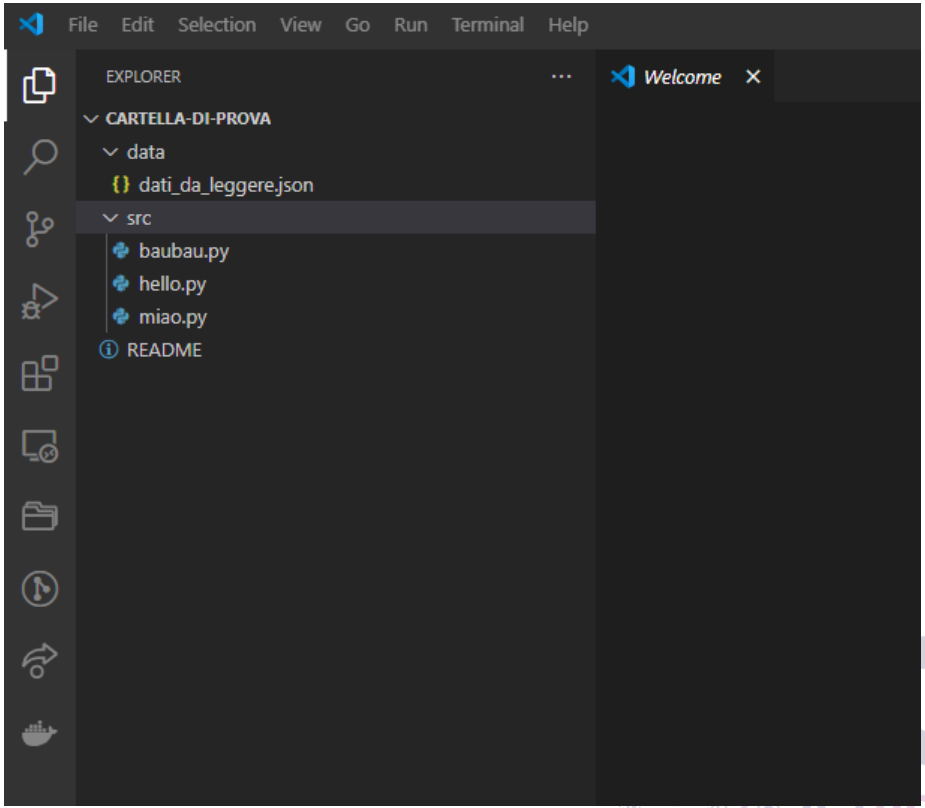
Visual Studio Code - Setup - Open the project

1. Click “**File > Open Folder**”
2. Choose the parent folder you just created
3. Open the file `hello.py` and write `print(“Hello World”)`



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Visual Studio Code - Setup - Running hello.py

1. Click “**Terminal > New terminal**”
2. Navigate to the parent folder of `hello.py`
3. Once there, write `python hello.py`

```
39348@DESKTOP-R63USTJ MINGW64 ~/OneDrive/Desktop/dblp/cartella-di-prova/src
$ python hello.py
Hello World
```

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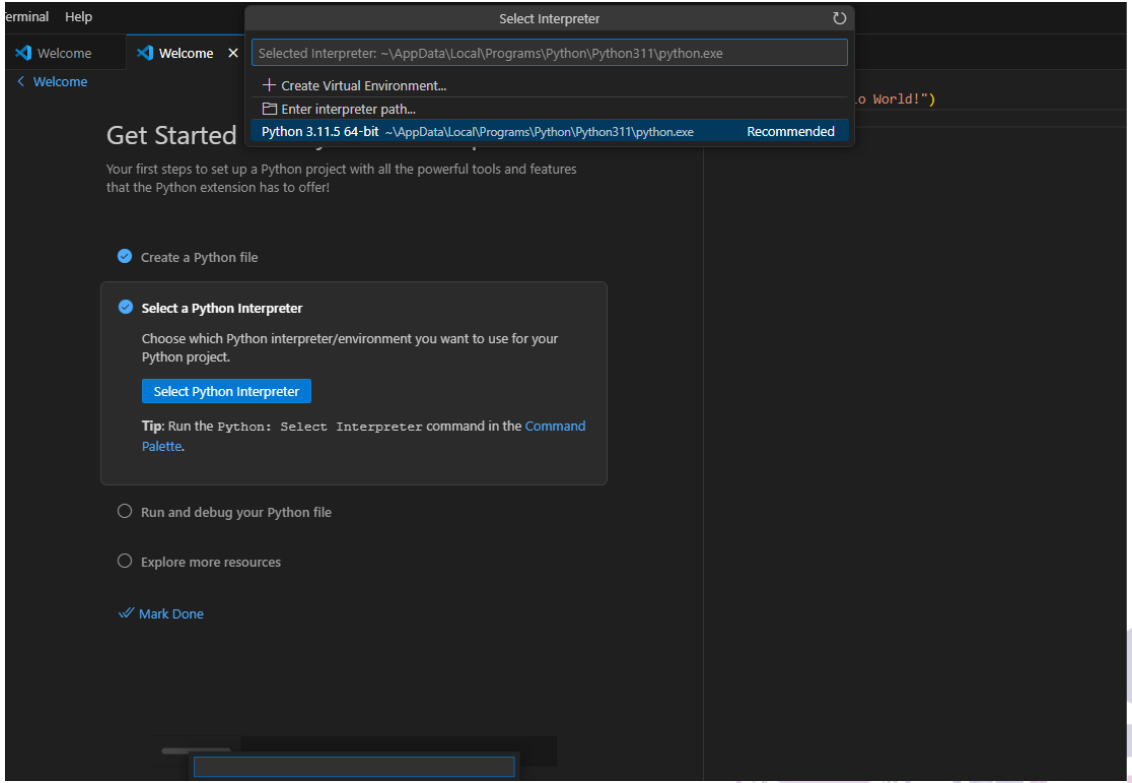
Visual Studio Code - Setup - Python environment

1. Click “**Create Python File**”
2. **Write** in file:
   `print(“Hello World!”)`
3. Go to “**Select** a Python **Interpreter**”
4. **Select** your python **version**.
5. **Go back** into the python **file**



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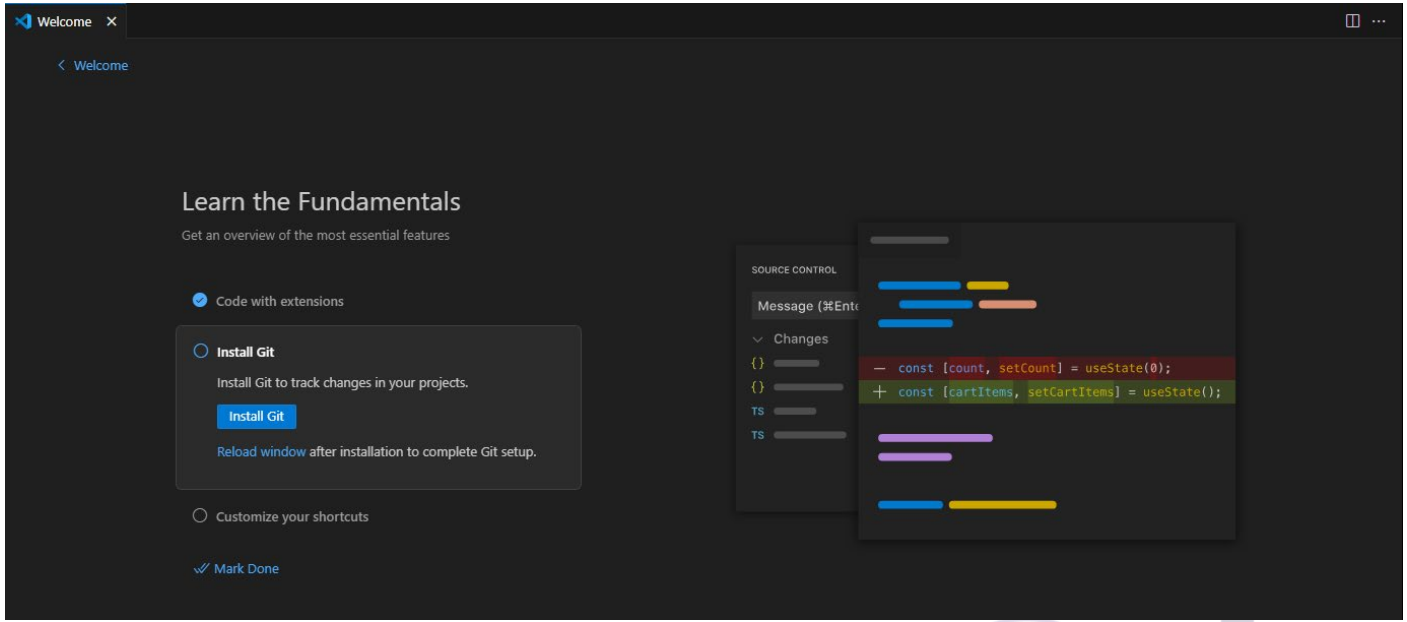
Visual Studio Code - Setup - Git Installation

1. Click “**Install Git**”
2. **Follow** the **instructions** on the pop-up Website.

A. **Download** & **execute** Installer
B. Use **default** settings.
C. **Use VScode** as “**default editor**”
D. Otherwise default options again (there are many)



3. Click “**Reload window**” in VScode

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